

Shaan Pakala

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[Google Scholar](#), [LinkedIn](#) & [GitHub](#)

About

I am a 1st year Computer Science Ph.D. student at the University of California, Riverside, where I work on machine learning AI4Science research problems with Professor [Vagelis Papalexakis](#). This includes developing interpretable surrogate models for designing new materials, or efficiently generating expensive physics simulation data using generative machine learning.

Education

- **Ph.D. in Computer Science** Sept. 2025 – Present
University of California, Riverside
Advised by Prof. Vagelis Papalexakis
- **B.S. in Data Science** Sept. 2021 – June 2025
University of California, Riverside

Research Experience

- **Lawrence Livermore National Laboratory** Summer 2026
Research Intern in the predictive healthcare group, with Dr. Priyadip Ray
- **Lawrence Livermore National Laboratory** Summer 2025
Graduate Research Intern in the predictive healthcare group, with Dr. Priyadip Ray and Dr. Braden Soper, working on causal inference for Parkinson's Disease progression
- **University of California, Riverside** June 2025 – Present
Graduate Student Researcher with Prof. Papalexakis, working on AI4Science problems using tensor methods
- **University of California, Riverside** June 2024 – June 2025
Undergraduate Researcher with Prof. Papalexakis

Awards

- **Travel Award** 2025
NeurIPS AI for Accelerated Material Design Workshop
- **Undergraduate Research Spotlight Award** 2025
Bourns College of Engineering – University of California, Riverside
- **Student Travel Award** 2024
IEEE International Conference on Big Data
- **NSF Research Experience for Undergraduates Fellowship** 2024
University of California, Riverside

Papers

Conference

- [1] **Tensor Methods: A Unified and Interpretable Approach for Material Design** 2026
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
ACM SIGKDD (AI for Sciences track) 2026 [[PDF](#)] [[Code](#)]
- [2] **Automating Data Science Pipelines with Tensor Completion** 2024
Shaan Pakala, B. Graw, D. Ahn, T. Dinh, M.T. Mahin, V. Tsotras, J. Chen, and E.E. Papalexakis
IEEE International Conference on Big Data 2024 [[Link](#)] [[PDF](#)] [[Code](#)]

Workshop

- [3] **Surrogate Modeling for the Design of Optimal Lattice Structures using Tensor Completion** 2025
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
NeurIPS AI for Accelerated Material Discovery Workshop 2025 [[PDF](#)] [[Code](#)]
- [4] **Efficiently Generating Multidimensional Calorimeter Data with Tensor Decomposition Parameterization** 2025
P. Goulart*, Shaan Pakala*, and E.E. Papalexakis
ICCV Representation Learning with Very Limited Resources Workshop 2025 [[PDF](#)] [[Code](#)]
- [5] **Tensor Completion for Surrogate Modeling of Material Property Prediction** 2025
Shaan Pakala, D. Ahn, and E.E. Papalexakis
AAAI Bridge on Knowledge-Guided Machine Learning 2025 [[PDF](#)]

* denotes equal contribution